

Stage 9 End of unit 12

The tests and mark schemes have been written by the authors. These may not fully reflect the approach of Cambridge Assessment International Education.

Answers

1 a 0.32 [1]

b 0.47 [1]

2 a $\frac{7}{16}$ [1]

b $\frac{6}{16}$ or $\frac{3}{8}$ [1]

3 a Any reasonable answer. For example: They travel to work together. [1]

b Any reasonable answer. For example: They work in different places. [1]

4 a If X happens the number is 2 or 4 and $P(Y) = 0.5$.
If X does not happen the number is 1 or 3 and $P(Y) = 0.5$ again. [2]

b If X happens the number is 2 or 4 and $P(Z) = 0.5$.
If X does not happen the number is 1 or 3 and $P(Z) = 1$, which is different.
(Allow any equivalent explanation.) (Award 1 mark for an incomplete answer.) [2]

5 a $\frac{3}{5}$ [1]

b $\frac{7}{20}$ [1]

6 a

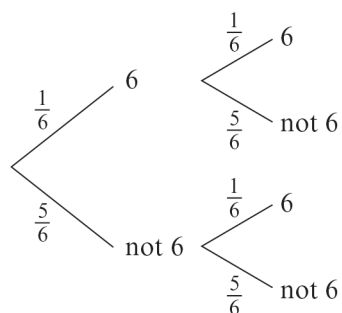
	First	Second	Outcome
$\frac{3}{4}$ B	$\frac{1}{3}$ B	B, B	
	$\frac{2}{3}$ W	B, W	
$\frac{1}{4}$ W	$\frac{1}{3}$ B	W, B	
	$\frac{2}{3}$ W	W, W	

[2]

b i $\frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$ [1]

ii $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ [1]

7 a First Second



[2]

b $1 - \frac{25}{36} = \frac{11}{36}$ or an equivalent method [2]

[Total: 20 marks]